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Behold! The 1-d wave equation!

$$\rho_0(x)\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial x}\left(T(x,t)\frac{\partial u}{\partial x}\right) + \rho_0(x)Q(x,t)$$

We get some variations of this equations making a few assumptions, if the tension does not change much in the string then $T(x,t) = constant$ so we could have:

$$\rho_0(x)\frac{\partial^2 u}{\partial t^2} = \left(T_0\frac{\partial^2 u}{\partial x^2}\right) + \rho_0(x)Q(x,t)$$

If there are no body forces or we let a sagged state of the string be it's equilibrium position we can ditch $Q(x,t)$ and obtain:

$$\rho_0(x)\frac{\partial^2 u}{\partial t^2} = T_0\frac{\partial^2 u}{\partial x^2}$$

If we divide by $\rho_0(x)$ we obtain the most commnly written form of this PDE

$$\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2} \qquad \text{where } c^2 = \frac{T_0}{\rho_o(x)}$$

What kinds of boundary and initial conditions can we have?

Common Boundary Conditions

Fixed ends: $u(0,t) = 0$ or $u(L,t) = 0$

Forced ends: $u(0,t) = f(t)$ or $u(L,t) = f(t)$

Common Initial Conditions

Initial shape of string: $u(x,0) = f(x)$

Initial velocity of string: $\frac{\partial u}{\partial t}(x,0) = g(x)$

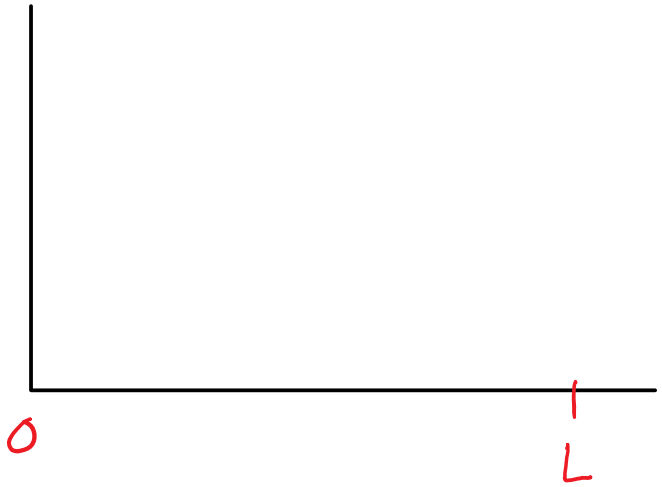
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Solving the wave equation

We are going to solve the wave equation assuming a uniform density $\rho_0(x) = \rho_0 = constant$ an constant tension T_0 and with two fixed ends a general initial shape and general initial velocity.

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \qquad \text{where } c^2 = \frac{T_0}{\rho_o(x)}$$

$u(0,t) = 0$ and $u(L,t) = 0$



Initial shape of string: $u(x,0) = f(x)$

Initial velocity of string: $\frac{\partial u}{\partial t}(x,0) = g(x)$

We begin the usual way by separating variables we assume: $u(x,t) = \phi(x)h(t)$

Think about what we might want our solution to be, we choose $\lambda > 0$ and attach a minus sign, that is we expect to have two ossilatory solutions here. It does turn out that for $\lambda = 0$ and $\lambda < 0$ we get only trivial solutions.

$$\frac{1}{c^2}\frac{1}{h}\frac{d^2h}{dt^2} = \frac{1}{\phi}\frac{d^2\phi}{dx^2} = -\lambda$$

Let's take a look at our equation for ϕ

$$\frac{1}{\phi}\frac{d^2\phi}{dx^2} = -\lambda$$

$$\phi(0) = 0 \quad and\ \phi(L) = 0$$

Any guesses for how this will play out?

Now let's look at our time dependent ODE for which we have no homogeneous boundary conditions.

$$\frac{1}{c^2}\frac{1}{h}\frac{d^2h}{dt^2} = -\lambda$$

So we have $\phi_n(x) = c_2 \sin\left(\frac{n\pi x}{L}\right)$ and $h_n(t) = c_3 \cos\left(\frac{cn\pi}{L}t\right) + c_4 \sin(\frac{cn\pi}{L}t)$

Thus taking the product of the solutions

$$u_n(x,t) = c_2 \sin\left(\frac{n\pi x}{L}\right)\left(c_3 \cos\left(\frac{cn\pi}{L}t\right) + c_4 \sin\left(\frac{cn\pi}{L}t\right)\right) = A_n \sin\left(\frac{n\pi x}{L}\right)\cos\left(\frac{cn\pi}{L}t\right) + B_n \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{cn\pi}{L}t\right)$$

Applying super position

$$u(x,t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)\cos\left(\frac{cn\pi}{L}t\right) + B_n \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{cn\pi}{L}t\right)$$

Now we can apply our initial conditions. $u(x,0) = f(x)$ $\frac{\partial u}{\partial t}(x,0) = g(x)$

So we have initial condition equations:

$$f(x) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)$$

$$g(x) = \sum_{n=1}^{\infty} B_n \frac{n\pi c}{L} \sin\left(\frac{n\pi x}{L}\right)$$

We then find our A_n and B_n in any of the usual ways……

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So we have our solution:

$$u(x,t)=\sum_{n=1}^{\infty}\sin\left(\frac{n\pi x}{L}\right)\left[A_n\cos\left(\frac{n\pi ct}{L}\right)+B_n\sin\left(\frac{n\pi ct}{L}\right)\right]$$

Where

$$A_n=\frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)dx \qquad B_n=\frac{2}{n\pi c}\int_0^L g(x)\sin\left(\frac{n\pi x}{L}\right)$$

You might recall that it is possible to write

$$A_n\cos(\omega t)+B_n\sin(\omega t)=\sqrt{A_n^2+B_n^2}\,\sin(\omega t-\phi)$$

For us, $\omega=\frac{n\pi c}{L}$

Note though that the amplitude of the wave is given by $\sqrt{A_n^2+B_n^2}$. If we are vibrating a string it might be apparent to you that the larger the amplitude the louder the sound.

Consider $B_n=\frac{2}{n\pi c}\int_0^L g(x)\sin\left(\frac{n\pi x}{L}\right)$ For what value of n would this number be the largest?

So then when B_n is at its largest we would have our largest amplitude $\sqrt{A_n^2+B_n^2}$ and for a vibrating string the sound that the corresponding frequency, $\frac{\pi c}{L}$ produces would be the most audiable. We call this the fundemental frequency!

One could then adjust which frequency is heard by adjusting c or L, Recall that $c=\sqrt{\frac{T_0}{\rho_0}}$ so the fundemental frequency is $\omega\frac{\pi\sqrt{T_0}}{L\sqrt{\rho_0}}$

Increase in tension = increase in frequency and a higher pitch Longer lengths = decrease in frequency and a lower pitch

Higher Density = decrease in frequency and a lower pitch , note on a guitar the lowest pitch string is the thickest and the highest pitch string is the thinnest.

Of course our solution is the sum of terms: $\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right)\left[A_n \cos\left(\frac{n\pi ct}{L}\right) + B_n \sin\left(\frac{n\pi ct}{L}\right)\right]$

Each n corresponds to a specific frequency, we call these the normal modes of vibration and our solution is a super position of all of them, it is just that the first mode has the largest amplitude.

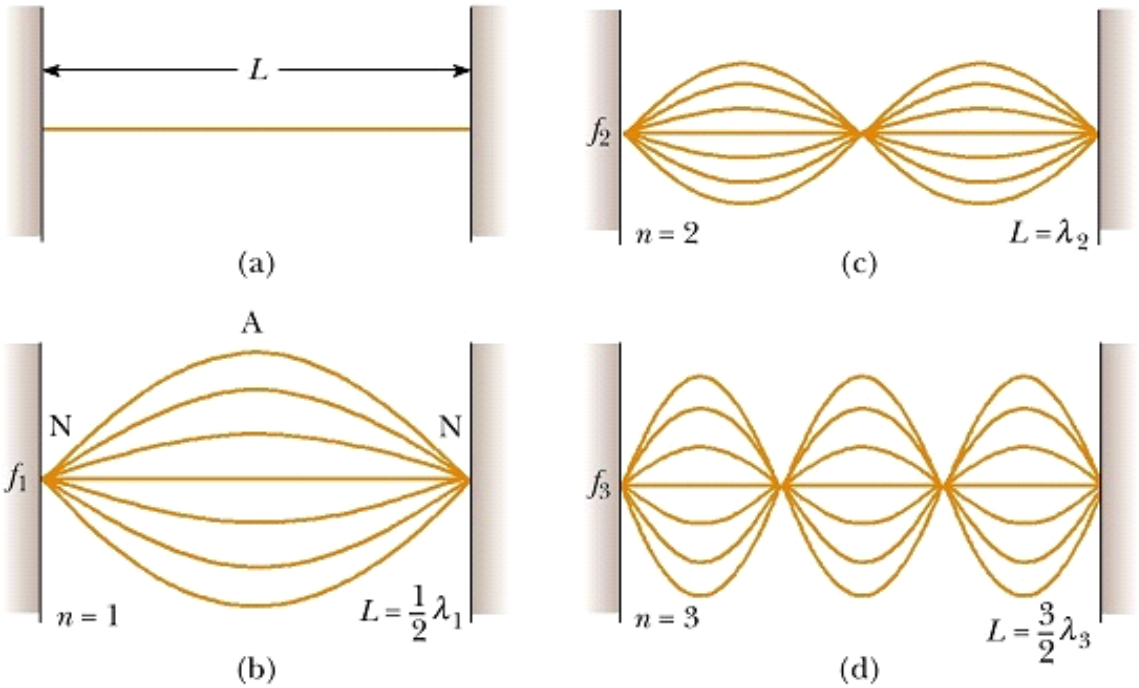


Figure 18.7 (a) A string of length L fixed at both ends. The normal modes of vibration form a harmonic series: (b) the fundamental, or first harmonic; (c) the second harmonic; (d) the third harmonic.

In the figure above each line is a normal mode plotted at a specific time, notice that the only thing that changes is the amplitude, also notice that there are places for $n > 1$ where the amplitude remains zero for all times t we call these nodes.

https://www.youtube.com/watch?v=6sgI7S_G-XI



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Homework Example: 15.4 #7

Suppose we have a vibrating string satisfying the following PDE problem.

$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ $\quad u(0,t) = 0$ and $u(L,t) = 0$ Initial shape of string: $u(x,0) = f(x)$

Initial velocity of string: $\frac{\partial u}{\partial t}(x,0) = 0$

Show that: $u(x,t) = \frac{1}{2}[F(x-ct) + F(x+ct)]$ Where F is the odd extension of $f(x)$

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A Damped String Homework Example 15.4 #3

Consider a damped string which satisfies.. $\rho_0 \frac{\partial^2 u}{\partial t} = T_0 \frac{\partial^2 u}{\partial x^2} - \beta \frac{\partial u}{\partial t}$

a) Why is Beta >0 to induce damping?

b)Determine the solution by separation of variables that satisfies the boundary conditions

$u(0,t) = 0$ and $u(L,t) = 0$ $u(x,0) = f(x)$ and $\frac{\partial u}{\partial x}(x,0) = g(x)$ We get to assume that $\beta^2 < \frac{4\pi^2 \rho_0}{L^2}$

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